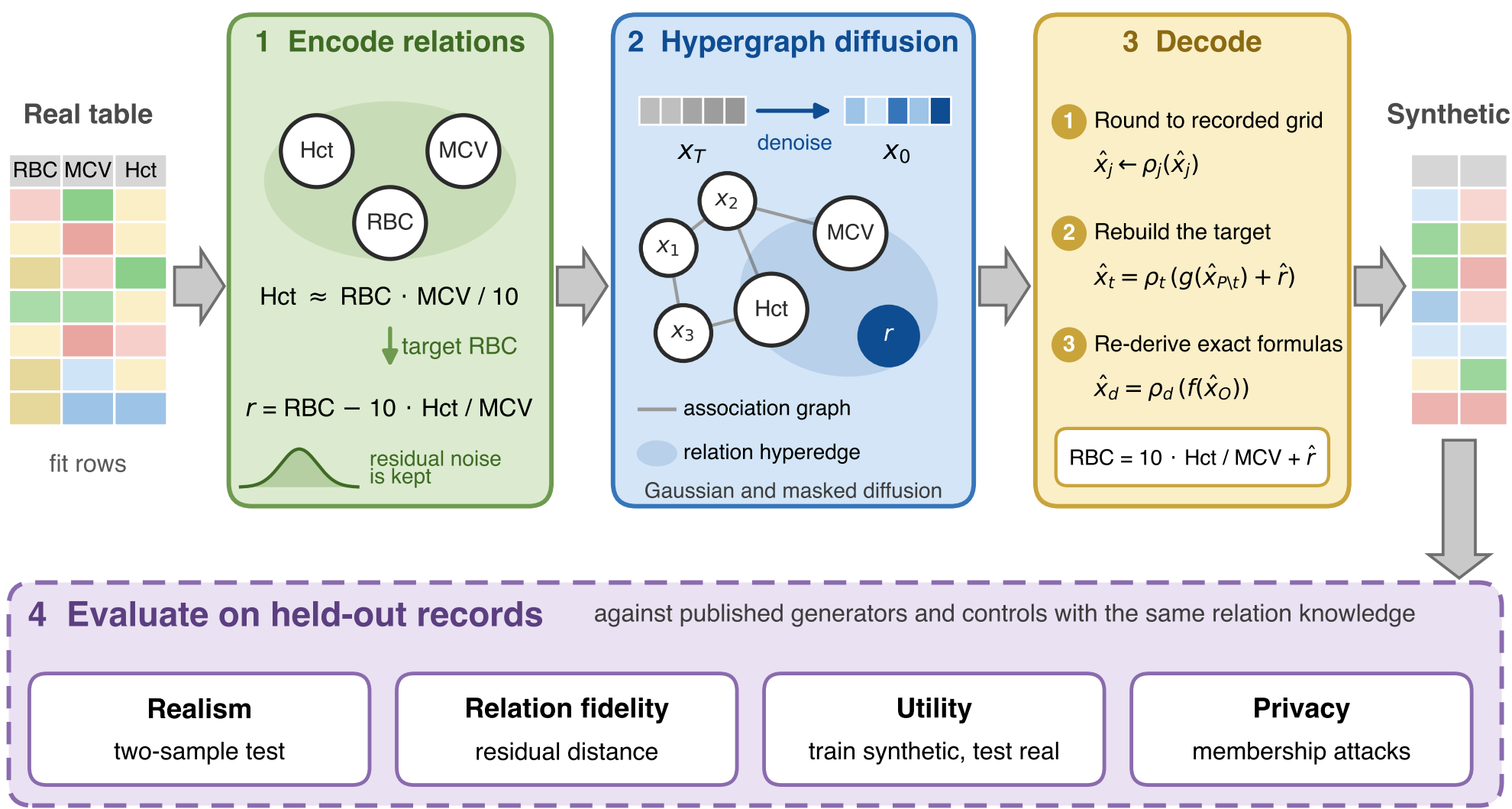
Real table
RBC MCV Hct
fit rows
1 Encode relations
Hct
MCV
RBC
Hct ≈ RBC · MCV / 10
target RBC
r = RBC − 10 · Hct / MCV
residual noise is kept
2 Hypergraph diffusion
x_T
denoise
x_0
x_1
x_2
x_3
MCV
Hct
r
association graph
relation hyperedge
Gaussian and masked diffusion
3 Decode
1 Round to recorded grid
$\hat{x}_j \leftarrow \rho_j(\hat{x}_j)$
2 Rebuild the target
$\hat{x}_t = \rho_t\,(g(\hat{x}_{P\setminus t}) + \hat{r})$
3 Re-derive exact formulas
$\hat{x}_d = \rho_d\,(f(\hat{x}_O))$
RBC = 10 · Hct / MCV + $\hat{r}$
Synthetic
4 Evaluate on held-out records
against published generators and controls with the same relation knowledge
Realism
two-sample test
Relation fidelity
residual distance
Utility
train synthetic, test real
Privacy
membership attacks